

WINDOWS FORENSICS / CTF WRITEUP

PaperGhost

A forensic walkthrough of nine findings

CASE ARC

A planted USB delivered a disguised update package. Registry, privacy-consent, resource-usage and search-index artifacts reconstruct execution, surveillance and data loss.

SYSTEM • NTUSER.DAT • SRUDB.DAT • WINDOWS.EDB

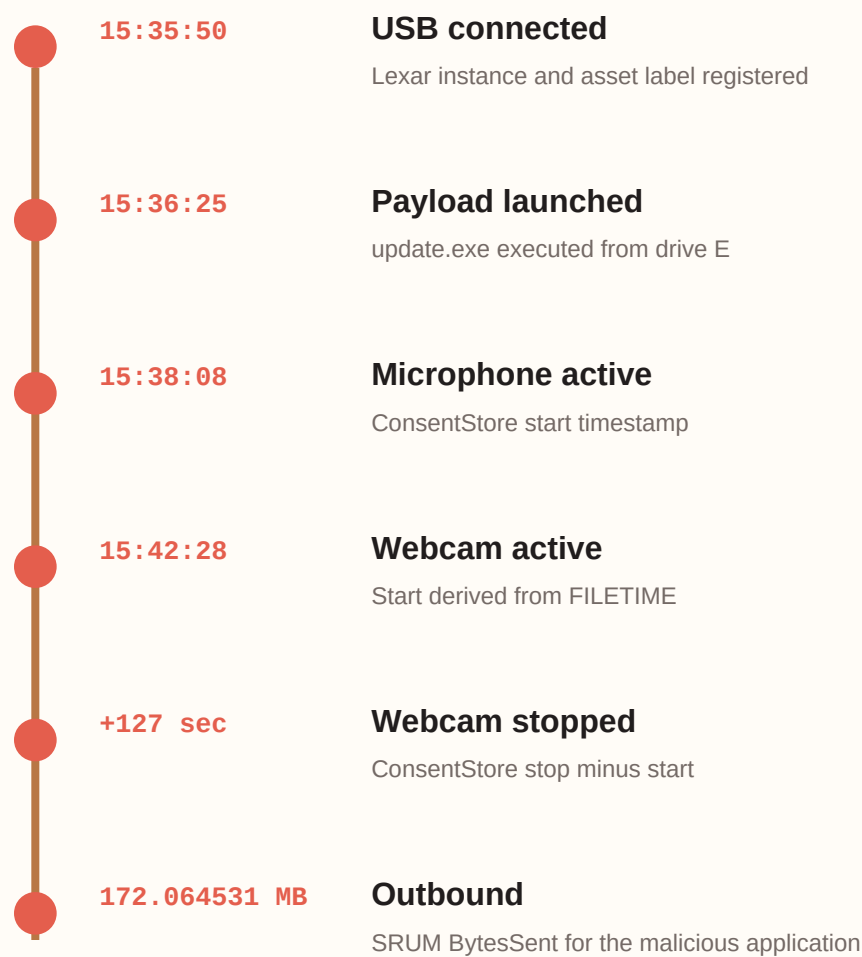
Prepared as a reproducible evidence walkthrough

Nine answers, four artifact families

The investigation connects removable-media arrival to payload execution, sensor access and network exfiltration. Each answer below is tied to a specific Windows artifact and a repeatable interpretation step.

1	USB connection time	SYSTEM registry hive
	2026-08-19 15:35:50	
2	USB serial number	SYSTEM registry hive
	RS200000000627E4&0	
3	Payload path	NTUSER.DAT UserAssist
	E:\C0-LT-0469 update package\update.exe	
4	Execution timestamp	NTUSER.DAT UserAssist
	2026-08-19 15:36:25	
5	USB asset name	SYSTEM registry hive
	C0-USB-0091	
6	Microphone capture start	NTUSER.DAT ConsentStore
	2026-08-19 15:38:08	
7	Webcam duration	NTUSER.DAT ConsentStore
	127	
8	Outbound traffic	SRUDB.dat
	172.064531	
9	Recovered credentials	Windows.edb search index
	tainsworth:D10g3n3s_T1ck3ts#2026	

Reconstructed activity sequence



INTERPRETATION

The sequence is internally consistent: initial access precedes execution, and execution precedes both microphone and webcam use. SRUM then records substantial outbound traffic attributed to the same payload.

FINDING 1

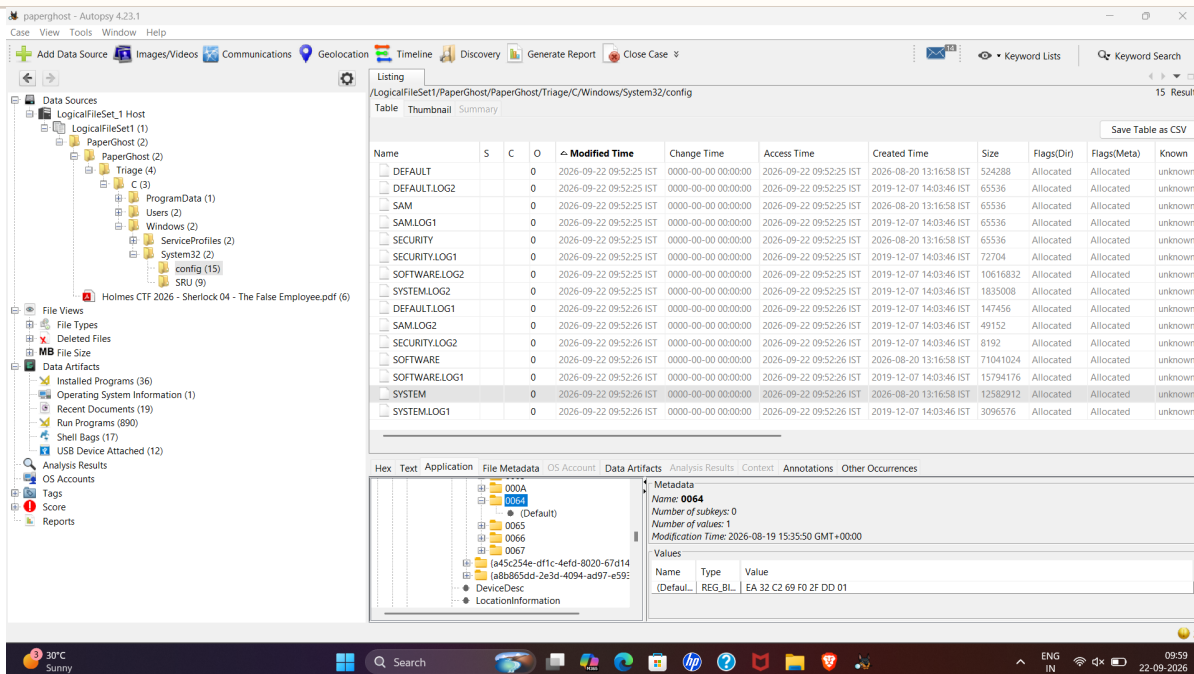
USB connection time

When did Clara Voss first connect the device Elias Venn left at her desk?

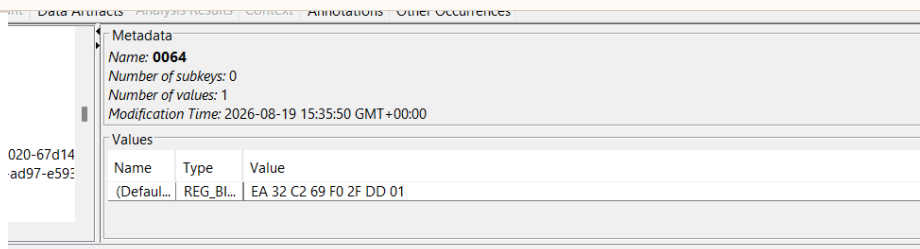
ANSWER

SYSTEM REGISTRY HIVE

2026-08-19 15:35:50



Source screenshot supplied from the Autopsy examination



Evidence detail

HOW THE EVIDENCE SUPPORTS THE ANSWER

The USB device instance contains a Device Properties timestamp under property 0064. The stored FILETIME converts to 2026-08-19 15:35:50.397412 UTC, so the challenge answer is rounded to whole seconds.

REPRODUCTION PATH

Autopsy: C:\Windows\System32\config\SYSTEM -> USB instance -> Properties -> 0064.

FINDING 2

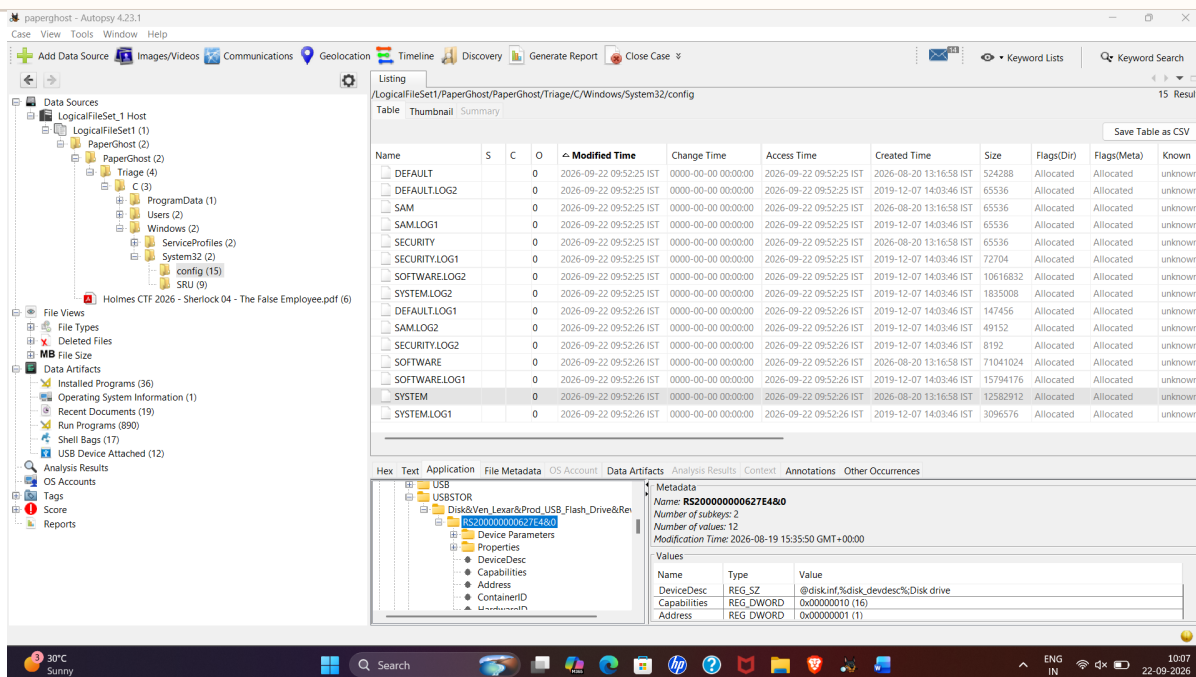
USB serial number

What serial number did the dropped device leave behind?

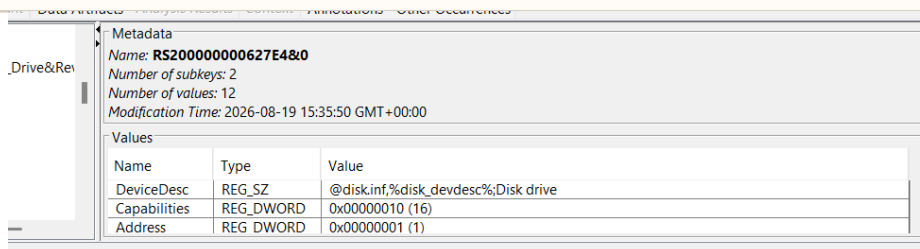
ANSWER

SYSTEM REGISTRY HIVE

RS200000000627E4&0



Source screenshot supplied from the Autopsy examination



Evidence detail

HOW THE EVIDENCE SUPPORTS THE ANSWER

USBSTOR identifies the Lexar device by its instance identifier. The complete instance component includes the '&0' suffix; omitting it produces an incomplete challenge value.

REPRODUCTION PATH

Autopsy: SYSTEM -> ControlSet001\Enum\USBSTOR -> Lexar USB Flash Drive instance.

FINDING 4

Execution timestamp

At what exact timestamp did Voss execute the malicious package?

ANSWER

NTUSER.DAT USERASSIST

2026-08-19 15:36:25

REPRODUCED ANALYSIS OUTPUT

```
Artifact : NTUSER.DAT / UserAssist / Count
Program  : E:\C0-LT-0469 update package\update.exe
FILETIME : 134316273857050000 (little-endian value)
Formula  : 1601-01-01 + FILETIME x 100 ns
UTC      : 2026-08-19 15:36:25.705
Answer   : 2026-08-19 15:36:25
```

HOW THE EVIDENCE SUPPORTS THE ANSWER

The UserAssist record embeds a Windows FILETIME. Interpreting the eight-byte value as little-endian and converting from the Windows epoch gives 2026-08-19 15:36:25.705 UTC.

REPRODUCTION PATH

Decode the UserAssist value name with ROT13, extract the embedded FILETIME, and convert it to UTC.

FINDING 5

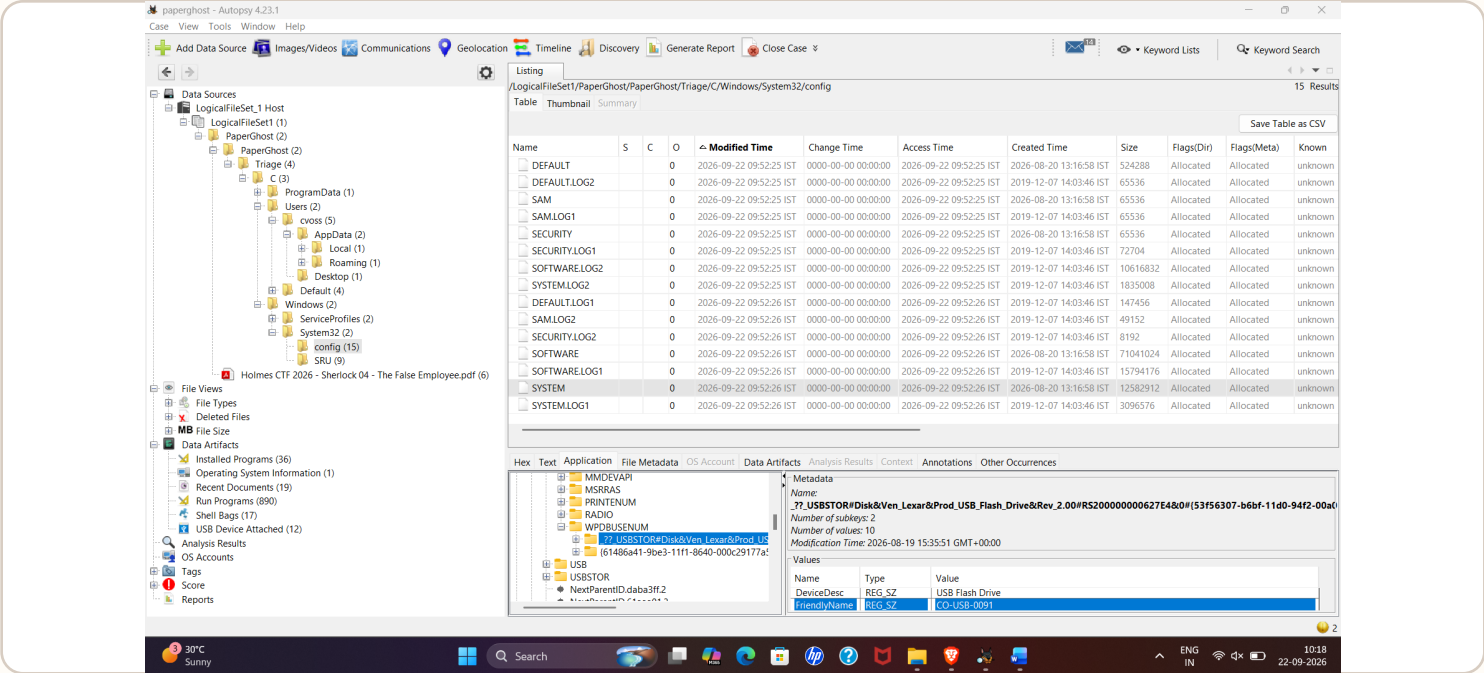
USB asset name

What device name surfaced when the DIOGENES-tagged USB connected?

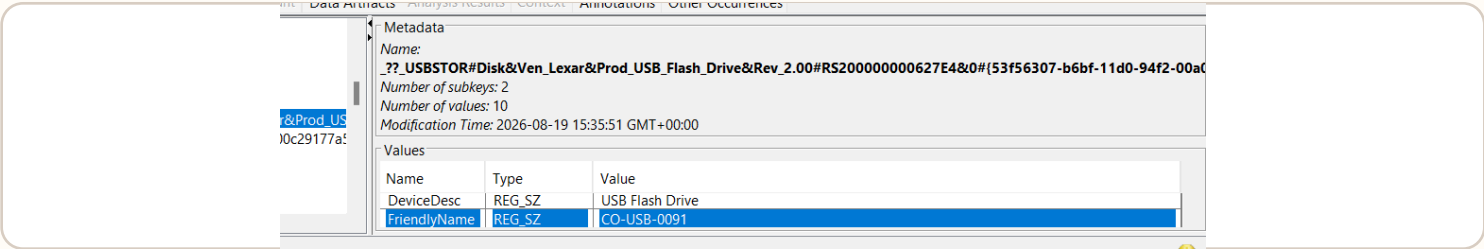
ANSWER

SYSTEM REGISTRY HIVE

CO-USB-0091



Source screenshot supplied from the Autopsy examination



Evidence detail

HOW THE EVIDENCE SUPPORTS THE ANSWER

The SWD WPDBUSENUM entry maps the USBSTOR instance to its portable-device metadata. FriendlyName contains the asset label CO-USB-0091.

REPRODUCTION PATH

Autopsy: SYSTEM -> ControlSet001\Enum\SWD\WPDBUSENUM -> FriendlyName.

FINDING 6

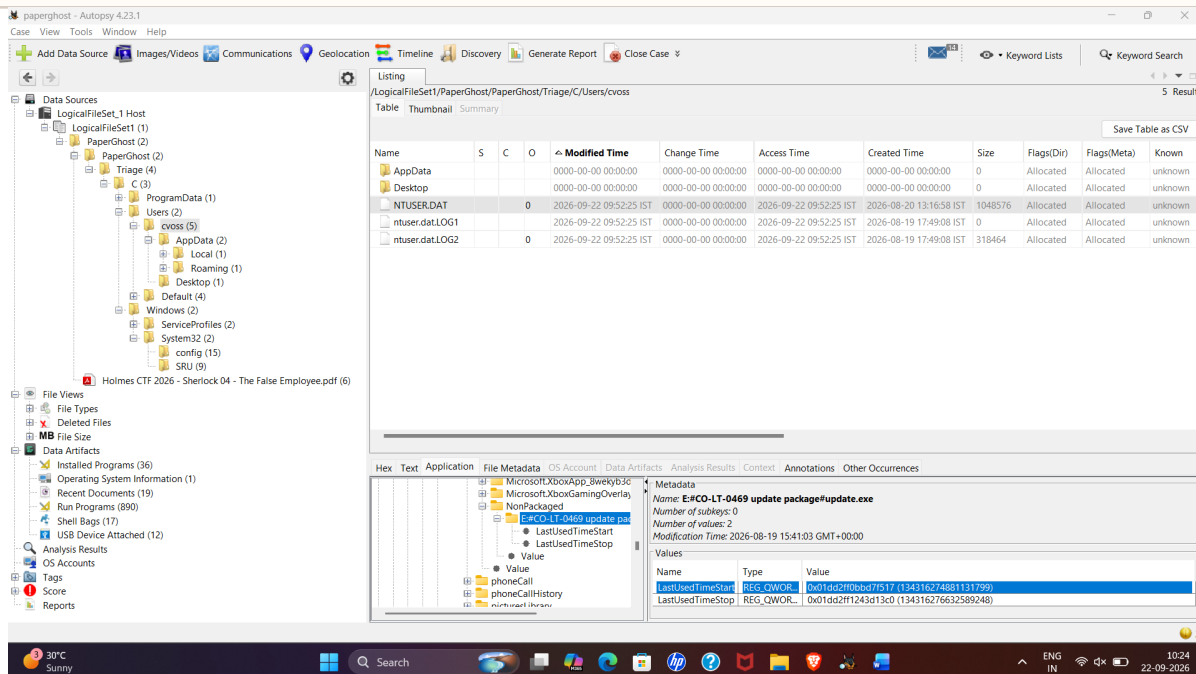
Microphone capture start

At what time did microphone capture begin?

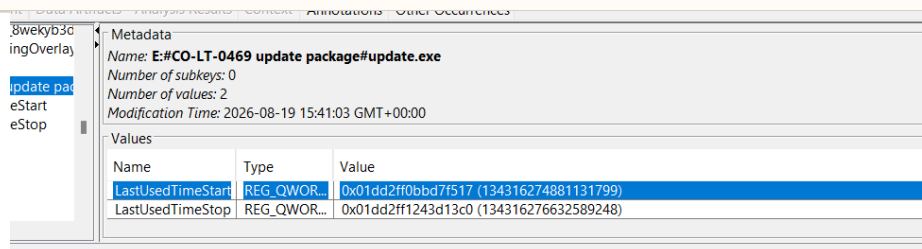
ANSWER

NTUSER.DAT CONSENTSTORE

2026-08-19 15:38:08



Source screenshot supplied from the Autopsy examination



Evidence detail

HOW THE EVIDENCE SUPPORTS THE ANSWER

The non-packaged update.exe entry under the microphone ConsentStore contains LastUsedTimeStart. Converting 134316274881131799 from Windows FILETIME gives 15:38:08.113179 UTC.

REPRODUCTION PATH

NTUSER.DAT -> CapabilityAccessManager\ConsentStore\microphone\NonPackaged\E:#CO-LT-0469 update package#update.exe.

FINDING 7

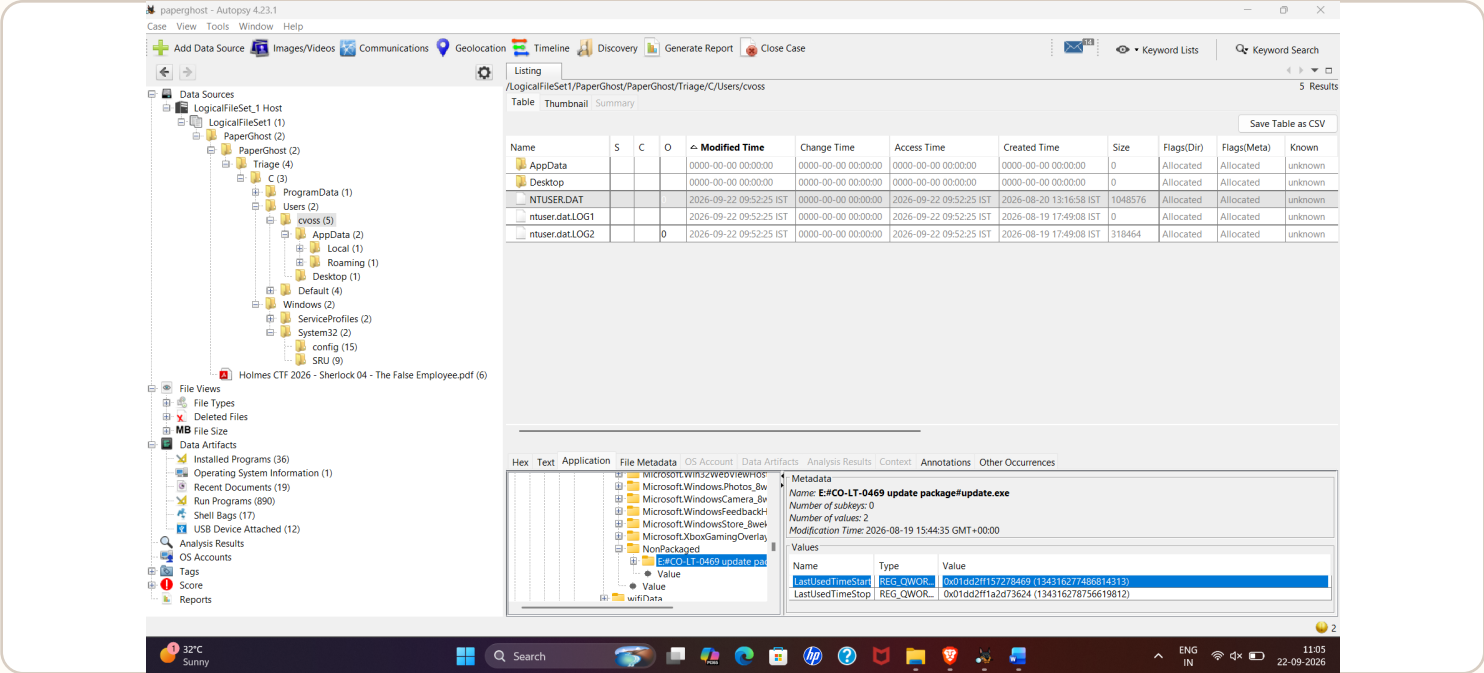
Webcam duration

For how many seconds did the webcam stream?

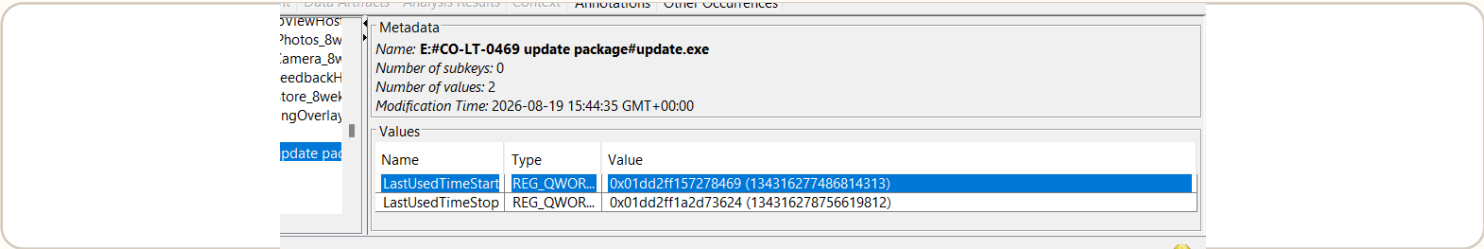
ANSWER

127

NTUSER.DAT CONSENTSTORE



Source screenshot supplied from the Autopsy examination



Evidence detail

HOW THE EVIDENCE SUPPORTS THE ANSWER

The webcam ConsentStore records a start of 134316277486814313 and stop of 134316278756619812. Their difference is 126.9805499 seconds, which rounds to 127 seconds.

REPRODUCTION PATH

NTUSER.DAT -> CapabilityAccessManager\ConsentStore\webcam\NonPackaged\E:#CO-LT-0469 update package#update.exe.

FINDING 8

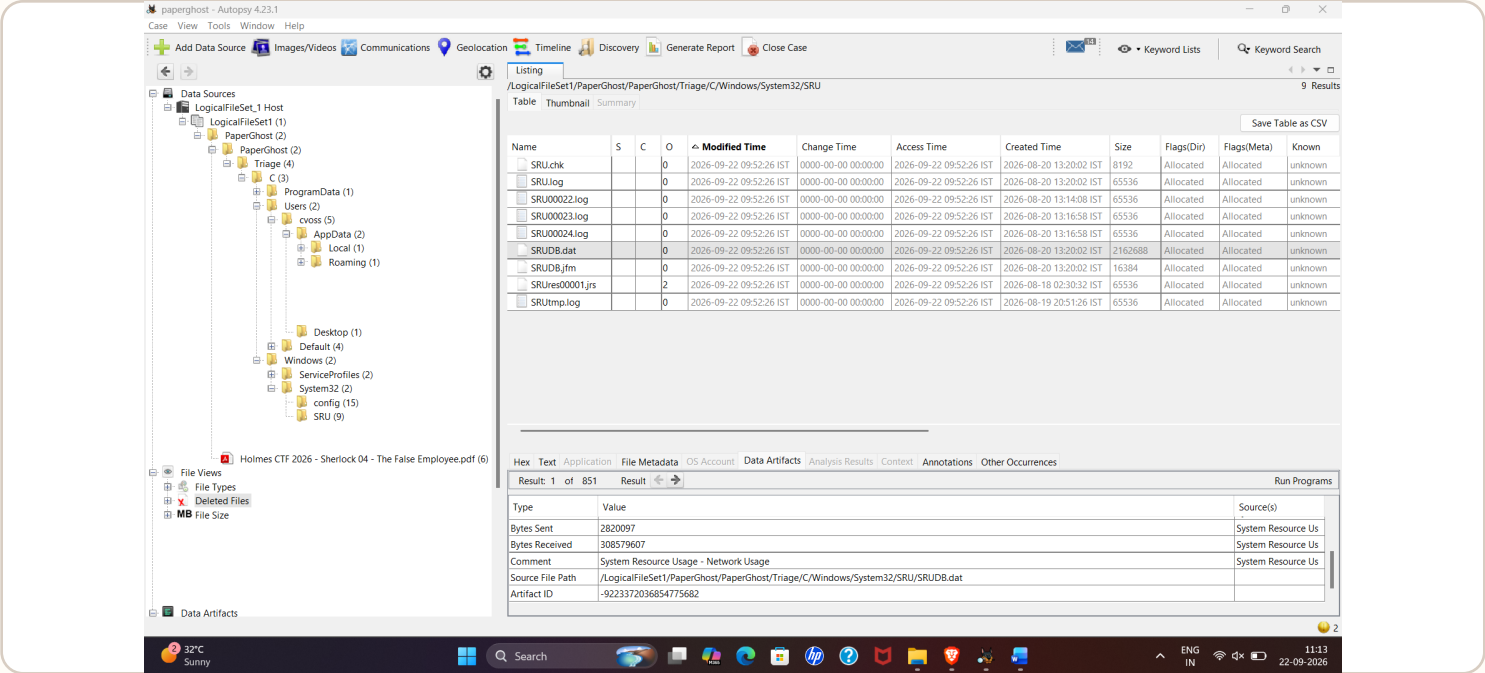
Outbound traffic

How many decimal megabytes flowed from the compromised machine to the C2?

ANSWER

172.064531

SRUDB.DAT



Source screenshot supplied from the Autopsy examination

Result: 1 of 851		Result	Run Programs
Type	Value		Source(s)
Bytes Sent	2820097		System Resource Us
Bytes Received	308579607		System Resource Us
Comment	System Resource Usage - Network Usage		System Resource Us
Source File Path	/LogicalFileSet1/PaperGhost/PaperGhost/Triage/C/Windows/System32/SRU/SRUDB.dat		
Artifact ID	-9223372036854775682		

Evidence detail

HOW THE EVIDENCE SUPPORTS THE ANSWER

The SRUM network table associates Appld 436 with update.exe and records 172,064,531 bytes sent. Decimal megabytes use 1,000,000 bytes per MB: $172,064,531 / 1,000,000 = 172.064531$ MB.

REPRODUCTION PATH

Parse SRUDB.dat; join the network usage record to Appld 436 and read BytesSent.

FINDING 9

Recovered credentials

What credentials were exposed for the DIOGENES ticketing developer?

ANSWER

WINDOWS.EDB SEARCH INDEX

tainsworth:D10g3n3s_T1ck3ts#2026

REPRODUCED ANALYSIS OUTPUT

```
Database : C:\ProgramData\Microsoft\Search\Data\Applications\Windows\Windows.edb
Table    : SystemIndex_PropertyStore
URL       : file:C:/Users/cvoss/Desktop/DIOGENES_26/EXT-0419.pdf
Name      : Tom Ainsworth
Title     : Software Developer - DIOGENES Ticketing Support
Username  : tainsworth
Password  : D10g3n3s_T1ck3ts#2026
Result    : tainsworth:D10g3n3s_T1ck3ts#2026
```

HOW THE EVIDENCE SUPPORTS THE ANSWER

Although the source PDF was absent, Windows Search retained indexed text for EXT-0419.pdf in SystemIndex_PropertyStore. The cached document text identifies Tom Ainsworth and preserves the username and password.

REPRODUCTION PATH

Export Windows.edb, enumerate SystemIndex_PropertyStore, and search record text for EXT-0419, DIOGENES, tainsworth or Password.

What the artifacts establish

Initial access

A Lexar USB identified as CO-USB-0091 first appeared at 15:35:50 UTC.

Execution

The malicious update.exe ran from drive E at 15:36:25 UTC.

Collection

The executable accessed the microphone and webcam; webcam activity lasted 127 seconds.

Exfiltration

SRUM recorded 172,064,531 outbound bytes for the malicious application.

Credential exposure

Windows Search retained developer credentials from a missing DIOGENES document.

Forensic note: derived timestamps and durations are conversions of raw registry FILETIME values. The credentials are recovered from indexed text, not from the original PDF itself.